

Continuous Distillation

PRE-LAB PLAN

Unit Operations Lab 5

3/18/2026

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1. Experimental Objective

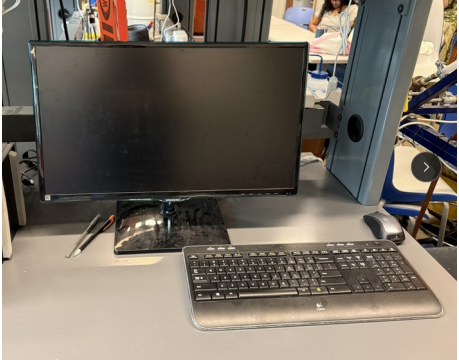
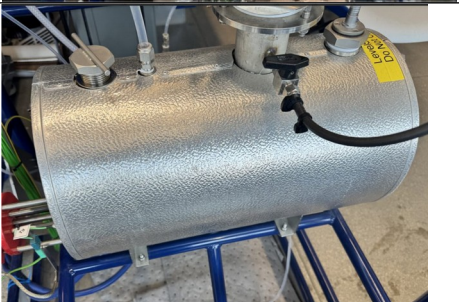
Using the Armfield UOP3CC Continuous Distillation Column for a 10 wt% ethanol reboiler solution and 20 wt% ethanol feed solution, the main objectives of this lab are to plot the temperature as a function of time under unsteady-state conditions, and to perform a McCabe-Thiele analysis under steady-state conditions using equilibrium data. The effect the feed stage has on composition will also be investigated.

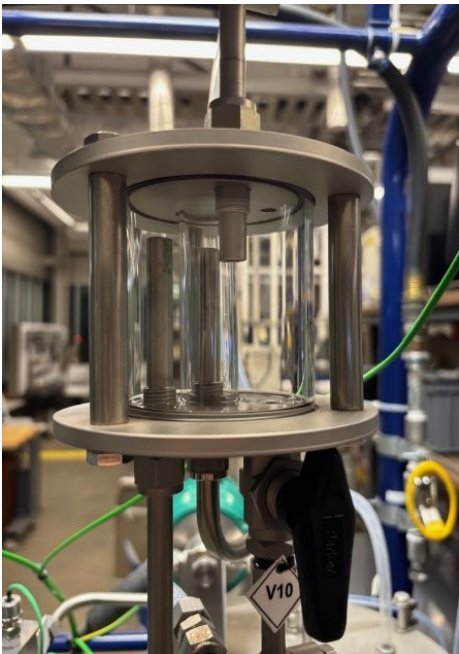
2. Experimental

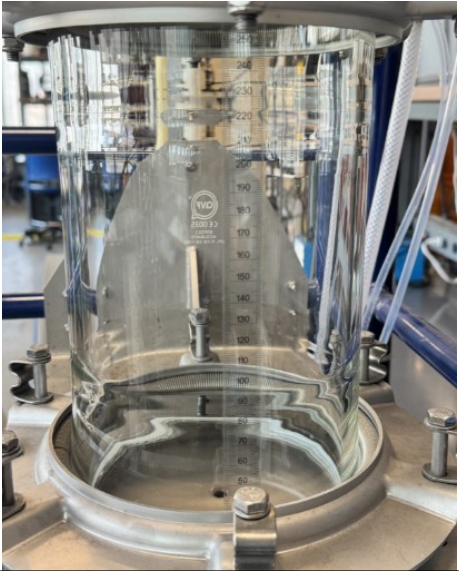
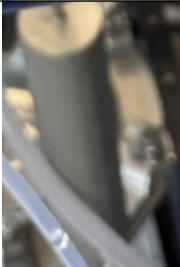
This lab is divided into three experiments: Experiment A involves calibrating the feed pump, Experiment B studies the effects of steady-state distillation of an ethanol-water mixture under continuous operation, and Experiment C investigates the effect of carrying the feed position under continuous operation. A summary of the apparatus and materials is provided below

2.1 Apparatus

<i>Number</i>	<i>Name</i>	<i>Main vs Auxiliary</i>	<i>Description</i>	<i>Image</i>
1	Armfield UOP3CC Continuous Distillation Column	Main	Primary unit for the continuous distillation of ethanol-water	
2	Control System/Computer	Main	System used to control column/reboiler specifications	

				
3	Condenser	Main	Condenses vapour overhead	
4	Reboiler	Main	Holds and heats reboiler solution	
5	Peristaltic Feed Pump	Main	Feeds liquid into system continuously	

6	Feed tank	Main	Holds feed solution before pumping	
7	Reflux system/Reflux regulator valve	Main	Controls reflux and top product split	
8	Accumulator	Main	Collects bottoms stream	

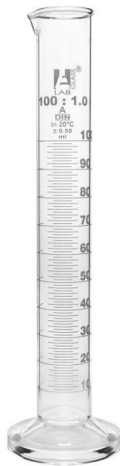
9	Top product receiver/Distillate collection	Main	Collects overhead product	
10	Heat exchanger (Bottoms)	Main	Used with cooling water to reduce overall temp. of bottoms product	
11	Drain Valve	Auxiliary	Allows for reboiler to be drained	

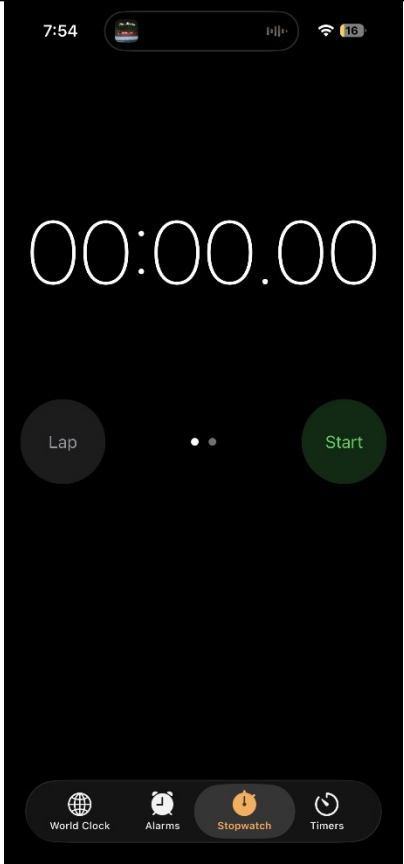
12	Water Flow Meter	Auxiliary	Monitors flowrate of water to condenser	
13	Pressure gauge	Auxiliary	Monitors pressure of the vapour present in the column	

2.2 Materials and Supplies

<i>Number</i>	<i>Name</i>	<i>Material vs Supply</i>	<i>Description</i>	<i>Image</i>
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1	Water	Material	Used to prepare ethanol-water mixtures	
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2	Ethanol	Material	Used to prepare ethanol-water mixtures	
3	Graduated cylinder	Supply	Used to collect product for testing	

4	Stopwatch	Supply	Used for timing	
5	Funnel	Supply	Used when loading solution	

6	SNAP 41 densitometer	Supply	Used to measure sample conc.	
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2.3 Experimental Plan

Experimental Variables:

Independent Variables: Feed position in the column, Feed flow rate, operating condition of the column of each run

Dependent Variables: Top product ethanol concentration, Bottom product ethanol concentration, tray temperatures, column efficiency, number of theoretical stages from McCabe-Thiele analysis

Controlled Variables: Reboiler solution composition, feed solution composition, reflux ratio, reboiler power, cooling water flow, sample timing

Setup:

1. Prepare the reboiler solution make a 10-liter of 10 wt% ethanol / 90 wt% water
2. Prepare the feed solution make a 3-liter solution of 20 wt% / 80 wt%
3. Check the system before starting up
 - a. Make sure all valves are closed and set the system to total reflux by switching off the reflux ratio timer and open valve V10 on the reflux pipe
4. Fill the reboiler by pouring the 10-Liter mixture into the reboiler
5. Fill the Feed bottle with the 3-Liter solution
6. Connect the blue cap and the black tubing to the feed pump
7. Install the feed pump tubing by opening the pump head clamp and insert the tubing between the rotor and pump head then close the clamp until it clicks shut

Experiment:

1. Turn on the control panel and start the condenser cooling water set the temperature to T9 and open valve V5
2. Adjust the cooling water flow to about 3L/min
3. Set the total reflux to 0.75kW, reflux ratio to 5:1, open V1 so the bottom product goes through the heat exchanger into the receiver, and allow the column to stabilize
 - a. Letting the columns stabilize can take up to 30 minutes
4. Feed the column about 400 mL
5. Take an overhead sample through valve 3 and take a bottom product sample through V2
6. Take 2 more samples of the top and bottom while spacing the samples about 5 minutes apart
7. Record temperatures through trays 1 and 9
8. Measure the sample concentration using the SNAP 41 densitometer
9. Turn off the feed pump, reflux switch, reboiler power, and close valve 1
10. Repeat the procedures for a different feed position
 - a. Disconnect the feed tube from the midpoint and connect it to the position above the top plate

Data Analysis:

1. Feed pump calibration data
 - a. Flow rate vs. Pump setting
2. Temperature vs. time data
3. Top product and bottom product composition
4. Tray temperatures
5. McCabe-Thiele analysis
6. Column efficiency
 - a. Compare the ideal McCabe-Thiel stage results to experimental values
7. Energy balance
 - a. Using the reboiler and condenser heats also discuss if the heat losses may affect the results

Safety Analysis:

Hazard	Yes/No	If Yes, Description
Potential electrical hazards?	Yes	Electric reboiler, peristaltic pump, electrical sensors present throughout system.
Potential mechanical hazards?	Yes	Rotating assembly of peristaltic pump presents pinch point.

Potential pressure hazards?	Yes	Vapor pressure from boiling liquid present in column
Potential temperature hazards?	Yes	Reboiler can reach high temperatures, possibly in excess of 100 °C
Hazardous/toxic chemicals involved	Yes	Ethanol is a skin irritant
Gloves required?	Yes	Ethanol is a skin irritant
MSDS reviewed by all team members	Yes	Ethanol(64-17-5), Water (7732-18-5)
Process parameters		
Temperature (°C)	100	
Pressure (psia)	40	
Safety controls	Yes	Reboiler has level sensor and will turn off if out of liquid, entire system is computer controlled.

I have read relevant background material for the Unit Operations Laboratory entitled: “Continuous Distillation” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

<u>Alexis Guillen</u>	<u>3/18/26</u>
Signature of team member	Date
<u>Khoa Nguyen</u>	<u>3/18/26</u>
Signature of team member	Date
<u>Rishabh S. Patel</u>	<u>3/18/26</u>
Signature of team member	Date

<u>Kyle Hankosky</u>	<u>3/18/26</u>
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<u>Miguel Piza</u>	<u>3/18/26</u>
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